

Kalori

An AI-first calorie and nutrition tracker, rendered as a printed ledger.

PROJECT REPORT • MVP REQUIREMENTS LOCKED • DA NANG • 2026-05-01

Kalori is a dark-only, single-user Progressive Web App that replaces global food-database search with three unified logging modes — natural-language parsing, photo recognition, and a personal food library that grows from everything the user logs. Built for premium visual craft and Vietnamese-food accuracy that mainstream trackers cannot deliver.

§ 01 What is Kalori?

Kalori is an **AI-first nutrition tracker** built around a Gemini Flash model that handles three duties behind server-only Route Handlers: *text parsing* ("two slices of toast and a coffee"), *photo vision* (a bowl of bún bò photographed at breakfast), and a *weekly review* that summarises seven days of eating into a short editorial brief. Every meal the user logs becomes a one-tap entry in a personal food library, so the app gets faster — not slower — with use.

The dashboard renders as a printed ledger: a masthead, a chronometer ring counting remaining kilocalories, a meals bulletin, a water gauge, and a micronutrient heatmap. The aesthetic is editorial dark — oxblood #8A2A1F on warm near-black #0E0A08, set in Newsreader serif with Inter and JetBrains Mono — chosen to feel like a daily broadsheet rather than a fitness app.

Primary user

A Da Nang-based AI engineer eating mostly Vietnamese cuisine. Phone (375 px) for meal logging during the day, laptop (1280+) for the weekly review on Sundays. Premium-SaaS visual standard (Linear, Arc, Robinhood).

What it explicitly is not

No social, no sharing, no barcode scanner, no exercise log, no notifications, no light mode, no native app, no English/Vietnamese toggle. MVP is single-user and invite-only; the architecture caps at 10–100 invited cohort.

"Vietnamese / Asian cuisine — bún bò, phở, cơm tấm, bánh mì, bún thịt nướng — must be a first-class citizen, not an exception."

§ 02 The Fourteen Functions

Each function below corresponds to a locked acceptance criterion in the project's PRD (§3.1–§3.14). Together they cover the entire Yes column of the MVP matrix.

3.1 ONBOARDING

8-Step Onboarding Wizard

Guided sign-up that collects sex, age, height, weight, target, activity level, dietary preferences, and a target-calorie computation using the Mifflin–St Jeor equation. Persists to a **profiles** row gated by RLS.

3.2 TEXT LOG

Natural-Language Meal Entry

Free-text input ("two slices of toast, peanut butter, a banana") sent to Gemini Flash via a server-only Route Handler. Returns kcal, macros, and a short reasoning string the user can audit before confirming.

3.3 PHOTO LOG

Camera / Gallery Vision Logging

Phone-camera or gallery image goes to Gemini Vision, which identifies the dish, estimates the portion, and returns full nutrition with reasoning. Optimised for Vietnamese dishes first.

3.5 CONFIRM

Confirmation Screen with Reasoning

Before saving, the user sees a "Why these numbers?" panel with the AI's portion / ingredient breakdown — no black-box estimates.

3.7 WATER

Water Tracker

Two quick-add buttons — +Glass (250 ml) and +Bottle (500 ml) — write to a dedicated `water_log` table; gauge updates optimistically and syncs through the offline outbox.

3.9 WEIGHT

Weight Log + Auto-Recalc

Weight entries automatically recalculate the Mifflin–St Jeor target, with explicit user notification when the target shifts. No silent recalculation.

3.11 UNDO

5-Second Undo Toast

Every destructive action (delete entry, delete weight, clear water) shows a 5-second undo toast that reverses the operation server-side via the offline outbox.

3.13 EXPORT

CSV + JSON Data Export

User can download every entry, every weight, every water log as CSV or JSON. No platform lock-in.

3.4 LIBRARY LOG

Personal Food Library

Every confirmed entry becomes a one-tap re-log in a searchable personal library. Replaces the global food-database paradigm; the library is the database.

3.6 DASHBOARD

The Ledger Dashboard

Chronometer ring (kcal remaining), macro bars, meals bulletin, water gauge, micronutrient heatmap, and a weekly insight strip — all rendered as a printed broadsheet on first paint.

3.8 PROGRESS

Progress View · Five Charts

Calorie deficit / surplus over time, weight trajectory, macro distribution, meal-time patterns, and a calendar heatmap of logging consistency — five chart sections on a single scrollable page.

3.10 OVERRIDE

Auto / Manual Target Override

User can override the auto-computed kcal target with a manual number; the override is sticky until explicitly cleared.

3.12 AI REVIEW

Weekly Gemini Review (7-day Cache)

Once per week, Gemini reads the last 7 days of entries and writes a short editorial review. Cached for the week so cost is bounded.

3.14 DELETE

Account Delete (Cascade)

A four-step typed-confirmation flow that cascades a delete through all eight tables via a service-role-only RPC, then signs the user out and shows a success banner on the landing page.

§ 03 Technology Stack

LAYER	TOOLS
Frontend	Next.js 16.2 (App Router · Turbopack · React Compiler v1.0) · React 19.2 · TypeScript 5.7 strict · Tailwind v4 · shadcn/ui · Radix primitives
Backend	Supabase (Postgres · Auth · Storage · RLS) via @supabase/ssr · 17 SQL migrations

LAYER	TOOLS
AI	Gemini Flash (text parse · vision · weekly review) called from server-only Route Handlers
State / data	zustand 5 · idb-keyval 6.2 (offline outbox) · zod 4.3 validation
PWA	Serwist 9.5 (esbuild-bundled externally; Next 16 + Turbopack lacks plugin support)
Monitoring	Sentry 10.49 (errors-only, by policy)
Test	Vitest 4.1 + happy-dom · Playwright 1.59 (e2e + axe + visual) · MSW 2.13 · fake-indexeddb · @testing-library/react
Hosting / CI	Vercel (region iad1) · GitHub Actions · Lighthouse CI · Husky + lint-staged

§ 04 Codebase at a Glance

METRIC	VALUE
Complexity tier	Complex — 7 planning artifacts · 5 phases · 26 + 35 task cards
Tracked files	785
Source lines (non-test)	40,337
Test lines	49,524
Test : source ratio	≈ 1.23 : 1
Migration SQL lines	1,387
TS / TSX source files	246
Test files	291
App Router pages / layouts / route handlers	12 / 2 / 23
UI components	55
Database migrations	17
TODO / FIXME / HACK in source	1
@ts-ignore / @ts-expect-error directives	0
any usage in app/lib/components	1 file

§ 05 High-Level Structure

app/	Next.js App Router
(app)/	Authenticated dashboard, log flow, library, progress
(auth)/	Login, callback shell
(marketing)/	Public landing
api/	23 route handlers (3 AI · 4 entries · 6 library · 3 export · ...)
auth/callback/	Supabase OAuth callback
offline/	Offline page
sw.ts	Service worker source

lib/	Domain logic
nutrition/ (10)	Macro / micro / Mifflin math
offline/ (8)	Outbox, network state, replay
ai/ (7)	Gemini parse / vision / weekly review
stores/ (7)	zustand stores (log flow, dashboard, etc.)
library/ (6)	Personal food library
auth/ (4)	Refresh interceptor (R1 anchor) + helpers
supabase/ (3)	Server / client / service factories
aggregations/, dashboard/, motion/, i18n/	
components/	UI components
charts/ (16)	Progress charts, weight trajectory, heatmap
dashboard/ (13)	Chronometer, macros, meals, water, micros
nav/ (11)	Sidebar, profile menu, mobile drawer
pwa/ (6)	Install prompt, update prompt, offline indicator
settings/ (4)	Account delete flow, export, profile
toast/ (3)	Undo toast + queue
tests/	unit · integration · components · e2e · axe · visual · rls · lighthouse
supabase/migrations/	17 SQL migrations (0001 → 0017)
scripts/	build-sw, bundle-budget, migration appliers, seed, icons
eslint-rules/	4 custom rules (architectural invariants)
Planning/	PRD, architecture, design-doc, ui-design, tasks, testing-strategy

§ 06 Current Status & Health

Phases 1 through 4 are closed. Phase 5 sub-tasks 5.1, 5.2, and 5.3 are closed. Task 5.4 (the project-closure sweep) is in progress: steps 1–10 are green, but step 11 — the manual production-smoke pass against <https://kalori-one.vercel.app> — surfaced eleven user-visible issues that triggered a focused MVP-Stabilization sprint (35 task cards across five phases A→E, approximately a three-week budget). The brain-storm pipeline for that sprint reached `artifacts_complete` on 2026-05-01 and is ready to begin on user command.

QUALITY GATE	LAST RESULT
Vitest unit + integration	1731 / 1731 passing
Branch coverage	70.85 % (≥ 70 % floor)
AI accuracy fixtures	30 / 30
Lighthouse mobile	≥ 0.91
Production migrations applied	17 of 17 to <code>kalori-prod</code>
Sentry unresolved (prod)	0